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PAPER_989: F_U_Bi Inside-to-Outside Buoyancy Mass Portion

Abstract

We derive $F_{U,Bi}$, the inside-to-outside buoyancy mass portion within the UQFF framework. Unlike F_{U,Bi_i} (the net acceleration felt by an object falling inward through buoyancy layers), $F_{U,Bi}$ quantifies the total vacuum energy contributing to outward mass transport from the SCm medium. The key equation is:

$$F_{U,Bi} = \rho_{SCm} \cdot V \cdot S_{26}^2 \cdot \frac{|U_b|}{|U_g| + |U_b|}$$

1. Core Derivation

The buoyancy ratio measures what fraction of the gravitational-buoyancy system is dominated by outward buoyancy:

$$\text{ratio} = \frac{|U_b|}{|U_g| + |U_b| + \epsilon}$$

where $\epsilon = 10^{-300}$ prevents division by zero.

Layer Functions

$$U_g = \sum_{i=1}^{26} \frac{GM}{r^2} \cdot [\text{SSq}] \cdot \frac{i}{26}$$

$$U_b = \sum_{i=1}^{26} \frac{GM}{r^2} \cdot e^{-[\text{SSq}] \cdot i/26} \cdot \beta_i$$

Vacuum Density

$$\rho_{SCm} = \rho_0 \cdot e^{\kappa t}, \quad \rho_0 = 10^{-10} \text{ kg/m}^3$$

2. Solar System Calibration

For $M = M_{\odot}$, $r = 1$ AU: - $F_{U,Bi} = 2.33 \times 10^{40}$ (total vacuum energy contribution) - ratio = 0.606 (buoyancy-dominant)

3. Physical Interpretation

$F_{U,Bi} > 0$ always (mass flows inside $\rightarrow$ outside), while $F_{U,Bi_i} < 0$ (net acceleration outside $\rightarrow$ inside). The two quantities form a complementary pair describing the complete buoyancy dynamics of the UQFF vacuum.

4. Implementation

File: fubi_inside_outside.py, class FUBiInsideOutsideCalc. CP4 class #573.